Missive nouns, polysemy and coercion

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1 Background and main questions

1.1 The distribution of noun related complementiser clauses (NCCs)

(1) a. the allegation/belief/evidence/fact/information/report/statement that Bilbo found the ring
b. #the book/brochure/newsletter/newspaper that Bilbo found the ring
c. #the house that Bilbo met Gandalf
d. #the event/celebration that Gandalf set off fireworks

The predicate modification analysis: (Bondarenko, 2022; Elliott, 2020; Kratzer, 2006; Moulton, 2009)

- NCCs select for nouns that denote 'contentful entities' (entities that have a propositional content).
- E.g., eventualities including events (*allegation*) and states (*belief*)
- Data such as (1b) are taken to indicate that *book*, *brochure*, *newsletter*, *newspaper* do not denote contentful entities.

1.2 Polysemous noun with an informational sense

- Evidence from e.g., copredication constructions and numeral constructions
- *book*-like nouns can denote both physical objects and informational entities, even simultaneously (7c)

(2) a. Tolkien wrote/published four books. (Informational reading)
b. Millions of Tolkien's books have been sold. (Physical entity reading)
c. Alex memorised and then hid the book/brochure/newsletter/newspaper.

The dot-type analysis: (Asher, 2011; Asher & Pustejovsky, 2006; Chatzikyriakidis & Luo, 2020; Pustejovsky, 1995)

- Entities in the domain of such nouns are of a complex *dot type*
- E.g., the dot type PHYS • INFO reflects both an informational and physical 'aspect'

1.3 The puzzle: missive nouns

Surprising data from preparatory work testing predictions for a polysemy-based analysis of the selectional restrictions of NCCs in English and Finnish.[1]

English

- *article*, *email*, and *letter* were attested in NCC constructions

(3) From the ukWaC corpus (Baroni et al., 2009)
a. That is until he gets a letter that his only living relative Aunt Sally has died.

[1] Work on Finnish was done jointly with Jesse Humaljoki.

b. You will also receive an automatic confirmation email that your request has been received.
c. There was no summer slump last year - I read a few articles that the supply for gold has dropped, but so is demand now.

Finnish[2]

- *artikkeli* ('article'), *kirje* ('letter'), *sähköposti* ('email'), but also *dokumentti* ('document')

(4) Kv-mediassa on ollut artikkeleita, että sota-alueilla on runsaasti
intnl-media.INE be be.PTCP article.PL.PAR COMP war-area.PL.ADE be plenty
lapsia, joista ei tiedetä mitään.
child.PL.PAR REL.PL.ELA NEG know.PASS anything
'There have been articles in the international media that there are many children in war zones about whom nothing is known.'

(5) Kesäkuussa hän sai kirjeen, että vuokrasopimus voidaan uusia, jos
summer.INE he/she/they get.3.PST letter.ACC COMP rental.contract can.PASS renew if
hän maksaa velkansa.
he/she/they pay debt.POSS
'In summer he/she/they got a letter that the rental contract can be renewed if they pay their debts.'

(6) Osallistujat allekirjoittavat dokumentin, että he ovat ymmärtäneet
participant.PL sign.3.PL document.ACC COMP they be.3.PL understand.PTCP
kilpailun säännöt ja vastaavat omasta terveydestään.
competition.GEN rule.PL and answer.3.PL own.ELA health.ELA.POSS
'Participants sign a/the document that they understand the rules of the competition and are responsible for their own health.'

For both the Finnish and English studies

- either zero or very few attested cases for nouns such as *book* and *newspaper*

Why are the data surprising?

- From the perspective of NCC research: *article*, *document*, *email*, and *letter* denote contentful entities.
- From the perspective of polysemy research, they pattern like *book*

(7) a. Alex wrote four different articles/letters/emails. (Informational reading)
b. Alex mistakenly sent me two identical articles/letters/emails. (Phys entity reading)
c. Alex memorised and then hid the article/letter/email.

- If physical entities that have content like *book* are not contentful entities in the requisite sense, they should not be expected to licence NCC constructions

Main questions

- If nouns such as *article*, *letter*, and *email* pattern semantically with *book*, why can they be used in NCC constructions?
- What are the most viable accounts (either semantic or pragmatic) that can account for these data?

[2] ACC = accusative case; ADE = adessive case; ELA = elative case; COMP = complementiser; GEN = genitive case; INE = inessive case; NEG = negation; PAR = partitive case; PASS = passive; PL = plural; POSS = possessive; PST = past; PTCP = participle; REL = relative pronoun.

1.4 Outline

- Further data collection (English) with a larger corpus
  - Looking for patterns in NCC constructions with missive nouns
  - A larger English corpus to check for NCC constructions with *book*-like nouns
- Outlining and assessing the predictions of possible analyses
- Presenting an analysis in terms of routinised coercion
- Discussion, problems and possibilities for further empirical testing

2 Data collection

2.1 Method

Noun set: *article*, *book*, *document*, *email*, *letter*, *newspaper*, *website*

- *article*, *letter*, *email* (and Finnish counterparts) found already by Sutton and Humaljoki.
- Adding *document* (*dokumentti* found in Finnish, but not in the smaller English corpus)
- *book*, *newspaper* (nouns frequently used as examples of polysemous nouns)
- *website* (added as a digital counterbalance to *email*)

Corpus: The EnTenTen21 corpus (Jakubíček et al., 2013): 52.2 billion words. POS tagged but not dependency parsed. Accessed via `sketchengine.eu`

A two-step data collection and refinement process:

1. Initial gathering of *N that* sentences
2. Dependency parsing, automatic data extraction and manual cleaning

Initial gathering of *N that* sentences

- Search parameter:
  - For *N* in: *article*, *book*, *document*, *email*, *letter*, *newspaper*, *website* any sentence containing *N that DP*.
- A random sample of 10k sentences per noun
- See Appendix A for more details

Dependency parsing, automatic data extraction and manual cleaning

- The 70k sentences were dependency parsed with the `en_trf` model of the NLP Spacy Python package (Honnibal & Montani, 2017).
- Python code was written to identify NCC constructions in the sentences for nouns in the noun set.
- Results manually checked (or, if more than 200 sentences per noun, a random sample of 200 sentences was checked)
- For examples of of manually excluded data, see Appendix A

2.2 Results

	article	book	document	email	letter	newspaper	website
Initial from corpus	10,000	10,000	10,000	10,000	10,000	10,000	10,000
After auto. extraction	175	18	141	784	304	10	8
Manually checked	175/175	18/18	141/141	200/784	200/304	10/10	8/8
NCCS (estimated)	150	2	107	(776)	(283)	3	0

- Many attested examples for *article*, *document*, *letter*, *email*
- Very few for *book* and *newspaper*, but these warrant further examination
- None attested for *website*, but this could be a data sparsity issue.

Examples of collected data

(8) *article*

a. I assume Rand knew that her life would be the subject of some controversy (there had already been articles that she had a "cult," etc.).

b. I read an article that the rivers and lakes around Tom Green county collected 3.8 billion gallons of water.

(9) *book* (the two attested examples from the corpus sample)

a. But Ehrman wrote a book that it is certain that Jesus has existed.

b. Trella signed her autograph book that she was "rich in a good man's love" signed Trella Evelyn White.

(10) *document*

a. She kept saying, 'You have to bring documents that I am Uighur.'

b. He could not bring any medical evidential documents that they were away at the time the deceased died.

(11) *email*

a. But then this morning we received another email that the flight was delayed.

b. I sent Sam an email that I was at the Convention Center.

(12) *letter*

a. The department has sent him a warning letter that he may be fired.

b. One Christmas, Cousins received a letter that their medical debt had been paid.

(13) *newspaper* (two of the three examples attested in the corpus sample)

a. I was sitting in San Francisco County jail that morning and read the newspaper that Fred Hampton and Mark Clark had been killed and murdered that morning.

b. They were shocked to read February 17 newspaper that the youth was nabbed from near the Doom bridge

2.3 Discussion

article, *document*, *letter*, *email*

- Likely that the proposition in the NCC is not the only content of the document
  - Perhaps the most salient or communicatively relevant content?

book, *newspaper*

- Two options:
  - Genuine but comparatively rare constructions

  - Result of typos or disfluency
- In either case, we need an account that in some way distinguishes *article*, *document*, *letter*, *email* from *book*, *newspaper* (and *website*)

3 Semantic assumptions and the space of possible analyses

3.1 The semantics of noun-NCC constructions

Following Bondarenko, 2022; Elliott, 2020; Kratzer, 2006; Moulton, 2009:

- NCCs express properties that modify nouns with Predicate Modification
- NCC clauses select for content-bearing entities/eventualities
  - Where content-bearing entities are subsort of generalised type e, e_c.
  - It is assumed that e subsumes v and some eventualities are of sort e_c

$$\begin{aligned}
[\![\text{that p}]\!] &= \lambda w.\lambda x_{e_c}.[\mathit{Cont}(x) = [\![\text{p}]\!]] \\
[\![\text{belief}]\!] &= \lambda w.\lambda x_{e_c}.[\mathit{belief}(x)] \\
[\![\text{claim}]\!] &= \lambda w.\lambda x_{e_c}.[\mathit{claim}(x)] \\
[\![\text{belief that p}]\!] &= \lambda w.\lambda x_{e_c}.[\mathit{belief}(x) \wedge [\mathit{Cont}(x) = [\![\text{p}]\!]]] \\
[\![\text{claim that p}]\!] &= \lambda w.\lambda x_{e_c}.[\mathit{claim}(x) \wedge [\mathit{Cont}(x) = [\![\text{p}]\!]]]
\end{aligned}$$

- *Cont* is a FUNCTION: for every entity, at most one proposition (Bondarenko, 2022; Elliott, 2020)

3.2 The semantics of *book*-like nouns

For today: suppress full details of a logic of polysemy

$$[\![\text{book}]\!] = \lambda w.\lambda x_{\sigma}.[\mathit{book}(x)]$$

Where σ could be a complex type such as PHYS • INFO

3.3 The space of possible analyses

- Type/sort distinctions between nouns
- Underspecification: Assignment of a free variable
- Coercion with free inference
  - constrained by context, saliency, background knowledge etc.
- Routinised coercion
  - constrained by lexical semantics

Next step: Outline and evaluate these four approaches

4 Type/sort distinctions between nouns

Nouns such as *article* and *letter* are of a different type or of a different sort to nouns such as *book* and *website*

- E.g., contentful entities of sort e_c and non-contentful entities of sort e_n

- NCC clauses select for the former and not the latter
- I.e. simply extend the class of contentful entities as needed

$$\begin{aligned}
[\![\text{that p}]\!] &= \lambda w.\lambda x_c.[\mathit{Cont}(x) = [\![\text{p}]\!]] \\
[\![\text{article}]\!] &= \lambda w.\lambda x_c.[\mathit{article}(x)] \\
[\![\text{book}]\!] &= \lambda w.\lambda x_n.[\mathit{book}(x)] \\
[\![\text{article that p}]\!] &= \lambda w.\lambda x_c.[\mathit{article}(x) \wedge [\mathit{Conts}(x) = [\![\text{p}]\!]]] \\
[\![\text{book that p}]\!] &= \text{SORTAL CLASH!!}
\end{aligned}$$

Predictions:

- *article/document/email/letter that p* is straightforwardly felicitous iff p is the contents of the article/letter/document/email/letter.

Objections:

- Doesn't seriously address the puzzle: Amounts to assigning types to nouns based on NCC data without accounting for the polysemy and copredication data
- Problem for the prediction: Incompatibility with *Cont* as a function
  Suppose you receive this email:

  > *Dear sir/madam,*
  > *Your library book is one week overdue. Overdue library loans incur fines and so you need to pay a fine of €2.*

  All of the following seem to be felicitous reports:

  (14) a. I got an email that I need to pay a fine (of €2).
  b. I got an email that my library book is overdue.
  c. I got an email that overdue library loans incur fines.

  This is incompatible with an analysis of *Cont* as a function.
  (It could be a relation $\mathit{Cont}(x, p)$ such that there need not be a unique p for every x).
- Problem for the prediction: *Cont* under-generates

  (15) He could not bring any medical evidential documents that they were away at the time the deceased died.

  - Such documents could be e.g., The results of a medical test done on a particular day.
  - Not obvious that the contents of such documents can be characterised as the proposition *x was not in location y on date z*
  - In this case at least, it would suffice to provide a document the contents of which implies/indicates that *x was not in location y on date z*

5 Underspecification: Assignment of a free variable

- Nouns such as *article* and *letter* are not of a different type to nouns such as *book* and *website*
- Revise the semantics for *that*-clauses to be underspecified
  - A variable that needs to be saturated in context in the sense of Recanati (2010)
  - For NCCs, instead of *Cont*, we have a free context-indexed variable Q_c

– Q must be resolved in context c to find an appropriate relation between the book/article and the relevant proposition:

$$\begin{aligned}
[\![\text{that p}]\!]^c &= \lambda w.\lambda x.[Q_c(x) = [\![\text{p}]\!]] \\
[\![\text{article that p}]\!]^c &= \lambda w.\lambda x.[\mathit{article}(x) \wedge Q_c(x) = [\![\text{p}]\!]] \\
[\![\text{book that p}]\!]^c &= \lambda w.\lambda x.[\mathit{book}(x) \wedge Q_c(x) = [\![\text{p}]\!]]
\end{aligned}$$

- This approach relies on context, saliency, background knowledge etc.
  - context allows finding an appropriate Q for *letter*, *email* etc.
  - context does not (normally) allow for finding an appropriate Q for *book*, *website* etc.
- Not proposed, to my knowledge, for these specific data

Predictions:

1. There are readings of e.g. *statement that p*, *belief that p* which have a reading other than that where p is the contents of the *belief*/*statement*
2. Priming of an appropriate Q should result in equally felicitous *article that p* and *book that p* constructions.
3. If Q_c were not a function, but a relation $Q_c(x, [\![\text{p}]\!])$ then e.g., *document that p* could be true for multiple p

Objections:

- Prediction 1 is hard to falsify, but I haven't found any verifiers for it
- Prediction 2 seems not to be borne out.

(16) Context: I found a some articles and books which made claims about the climate change debate.
  a. For instance, I found an article that climate change is worsening faster than expected.
  b. ?For instance, I found a book that climate change is worsening faster than expected.

6 Coercion: free inference vs. routinised coercion

- Nouns such as *article* and *letter* are NOT of a different type/sort to nouns such as *book* and website *website*: they denote individual objects
- NCCs select for nouns that denote content-bearing entities.
- There is a type clash that needs to be resolved between the noun and the NCC clause for nouns such as *article* and *letter* and nouns such as *book* and *website*

$$\begin{aligned}
[\![\text{that p}]\!] &= \lambda w.\lambda e_v.[\mathit{Conts}(e) = [\![\text{p}]\!]] \\
[\![\text{article}]\!] &= \lambda w.\lambda x_e.[\mathit{article}(x)] \\
[\![\text{book}]\!] &= \lambda w.\lambda x_e.[\mathit{book}(x)] \\
[\![\text{article that p}]\!] &= \text{TYPE/SORT CLASH!!} \\
[\![\text{book that p}]\!] &= \text{TYPE/SORT CLASH!!}
\end{aligned}$$

There are at least two ways of analysing such type clashes: free inference and routinised coercion

6.1 Free inference (constrained by saliency, background knowledge etc.)

- For some concept/property Q, insert Q at some point of the derivation to resolve the type clash.
- Example from the count/mass literature: *two white wines* interpreted in context as *two* GLASSES/BOTTLES/CARAFES OF *white wine*
- Complement coercion for some nouns:

(17) a. ?Lizzy enjoyed the rock. (Asher, 2011)
b. ?Alex started a chair yesterday.

(18) Context: Alex is a furniture maker.
a. She started a chair yesterday $\mapsto_{\text{coercion}}$ Alex started making a chair yesterday.

(19) (Bott, 2017)
a. Half a year ago, Peter started to jog four kilometers every day. When he began, he was quite slow but now he is really fast.
b. Yesterday, Peter jogged in only thirty minutes.

- A coercion-based analysis with free inference:

$$[\![\text{article that p}]\!] \mapsto_{\text{coercion}} Q([\![\text{article}]\!], [\![\text{that p}]\!]) \mapsto_{\text{coercion}} \begin{cases} \mathit{Say}([\![\text{article}]\!], [\![\text{that p}]\!]) \\ \mathit{Convey}([\![\text{article}]\!], [\![\text{that p}]\!]) \\ \ldots \end{cases}$$

Predictions:

1. an article, letter etc. could have more than one proposition as contents depending on how relations such as *Say*, *Convey* etc. are worked out
   - $[\![\text{that p}]\!] = \lambda x.\lambda p.[\mathit{Cont}(x) = p$, but now the relevant x is an eventuality
2. Priming of an appropriate Q should result in equally felicitous *letter that p* and *book that p* constructions.
   - I.e. the same as the second prediction for the underspecification/free variable account
   - Reason to also reject this approach

6.2 Routinised coercion (constrained by lexical semantics)

- A limited number of properties Q are listed as part of the lexical semantics of common nouns
- E.g., richer lexical semantic representations such as Generative Lexicon (Pustejovsky, 1995) (but alternatives include frames e.g, (Kallmeyer et al., 2017))
  - The Telic Qualia Field for *book* (roughly what books are for) contains *Read*
- Access Q from the lexical semantics of the noun to resolve the type clash.
- Suppose that the Telic Qualia field for *article* and *email* is not *Read*, but *Convey*, *Communicate* (or some similar predicate of eventualities)
- Basic idea (details to be worked out shortly):

$$[\![\text{article that p}]\!] \mapsto_{\text{coercion}} \lambda x.\mathit{article}(x) \wedge \mathit{Convey}(x, [\![\text{that p}]\!])$$

Does not make the false predictions of either of the previous two context-based approaches:

1. Not: There are readings of e.g. *statement that p*, *belief that p* which have a reading other than that where *p* is the contents of the *belief*/*statement*
2. Not: Priming of an appropriate *Q* should result in equally felicitous *letter that p* and *book that p* constructions.

Predictions:

1. Nouns such as *book* should be infelicitous with NCCs
   - E.g. ⟦book⟧ does not specify a content-bearing set of eventualities such as *Convey* or *Communicate* as part of its lexical semantics
   - Therefore nothing is routinely derivable from ⟦book⟧ that can combine with NCCs

A general worry:

- The success of this proposal turns on systematically identifying related sets of eventualities in the lexical semantics of nouns

7 Summary of analyses and predictions

	missive nouns have unique contents	Relation for *belief that p* from context	Context easily facilitates *book that p*
Type/sort distinction	✓	✗	✗
Underspecification/free variable	?	✓	✓
Coercion via free inference	?	✗	✓
Routinised coercion	?	✗	✗
Observed	✗	✗	✗

8 Developing a routinised coercion proposal

8.1 Background

- Richer lexical semantic representations such as Generative Lexicon (Pustejovsky, 1995)
- Many common nouns make a available a set of eventualities in their lexical semantics even if they do not denote eventualities
  - E.g., ⟦book⟧ in Pustejovsky, 1995 makes available a set of book-reading and a set of book-writing eventualities.

Simplified representation, ignoring the polysemy of *book*.

- I am making explicit the restriction on *read* and *write* to books

(20) a. $$[\![\text{book}]\!] = \begin{bmatrix} \text{ARG_STR} & = \lambda x.book(x) \\ \text{TELIC} & = \lambda x.\lambda y.\lambda e.read(e,x,y) \wedge book(x) \\ \text{AGENTIVE} & = \lambda x.\lambda y.\lambda e.write(e,x,y) \wedge book(x) \end{bmatrix}$$

b. Such that:
   (i) $\textsc{arg_str}(\llbracket\text{book}\rrbracket) = \lambda x[book(x)]$
   (ii) $\textsc{telic}(\llbracket\text{book}\rrbracket) = \lambda x.\lambda y.\lambda e.read(e,x,y) \wedge book(x)$
   (iii) $\textsc{agent}(\llbracket\text{book}\rrbracket) = \lambda x.\lambda y.\lambda e.write(e,x,y) \wedge book(x)$

Cases of complement coercion (e.g., *start the book*) are resolved via access to the predicates of eventualities in the qualia fields of the nouns:

- As a subject control verb, ⟦start⟧ can bind an agent argument:

(21) $\llbracket\text{start}\rrbracket = \lambda P_{\langle e,\langle v,t\rangle\rangle}.\lambda x.\lambda e.start(e,P(x),x)$

- I assume that ι binds x in all qualia fields:

(22) $$\llbracket\text{the book}\rrbracket = \begin{bmatrix} \textsc{arg_str} & = \iota x.book(x) \\ \textsc{telic} & = \lambda y.\lambda e.\iota x.read(e,x,y) \wedge book(x) \\ \textsc{agentive} & = \lambda y.\lambda e'.\iota x.write(e',x,y) \wedge book(x) \end{bmatrix}$$

- On the telic interpretation, ⟦start the book⟧ $\mapsto_{\text{coercion}}$ ⟦start to read the book⟧

(23) $$\begin{aligned} \llbracket\text{start the book}\rrbracket &\mapsto_{\text{coercion}} \llbracket\text{start}\rrbracket(\textsc{telic}(\llbracket\text{the book}\rrbracket)) \\ &\mapsto_{\text{coercion}} \lambda y.\lambda e.start(e,[\lambda e'\iota x.read(e',x,y) \wedge book(x)],y) \end{aligned}$$

8.2 Application to missive nouns

Outline

- *book* and *website* make available a set of eventualities (e.g., *read*) that relate an entity (e.g., a book) to an agent
- *article* and *letter* are different
  - they make a available a different set of CONTENT BEARING eventualities e.g., *convey*
- The reading of e.g., *a letter that p* is approximately:
  'a letter that is intended to convey that p'

Verbal *convey that p* constructions: assumptions

- the *agent* argument is not represented, since *convey* does not mandatorily take an agent (*the letter conveyed the news*)
- Following e.g., Bondarenko, 2022; Elliott, 2020; Kratzer, 2006; Moulton, 2009:
  - the NCC is not an argument of the verbal predicate
  - composes via predicate modification (*PM*) which can be defined as a relation between predicates

(24) $PM := \lambda P.\lambda Q.P(x) \wedge Q(x)$

(25) a. $\llbracket\text{inform}\rrbracket = \lambda e.convey(e)$
   b. $\llbracket\text{that p}\rrbracket = \lambda e.[CONT(e) = p]$
   c. $$\begin{aligned} \llbracket\text{convey that P}\rrbracket &= PM(\llbracket\text{convey}\rrbracket)(\llbracket\text{that p}\rrbracket) \\ &= \lambda e.[convey(e) \wedge [CONT(e) = p]] \end{aligned}$$

For *the letter conveyed that p*, I assume that the *CAUSE* thematic role relation can be introduced in a similar manner as *AGENT* (Kratzer, 1996)

(26) $\theta_{\text{cause}} = \lambda x.\lambda e.CAUSE(e,x)$

This can combine with an expression of type $\langle v,t\rangle$ via *Event Identification* (Kratzer, 1996), or, more generally, Restrict, RT (Chung & Ladusaw, 2003):

- RT composes a relation of type $\langle\sigma,\langle\tau,t\rangle\rangle$ with a predicate of type $\langle\tau,t\rangle$ to yield a relation of type $\langle\sigma,\langle\tau,t\rangle\rangle$

(27) $RT := \lambda R_{\langle\sigma,\langle\tau,t\rangle\rangle}.\lambda P_{\langle\tau,t\rangle}\lambda x_{\sigma}.\lambda y_{\tau}.[R(x)(y)\wedge P(x)]$

Simplifying ⟦the letter⟧ to $\iota x.letter(x)$

(28) ⟦the letter conveyed that p⟧
$= RT((\theta_{\text{cause}})(\llbracket\text{convey that p}\rrbracket))(\llbracket\text{the letter}\rrbracket)$
$= \lambda e.[convey(e)\wedge[CONT(e)=p]\wedge CAUSE(e,\iota x.letter(x))]$

letter that p constructions via routinised coercion: assumptions

- In the context of ⟦letter⟧, letters are specified as that which conveys/communicates
  - Evidence for *convey* (or *communicate*) for *letter* and *email*, but not *book*:

(29) a. Alex wrote a letter/an email in order to convey her support.
b. ?Alex wrote a book in order to convey her thoughts.

(30) communicate via/with emails/letters/?books

- *convey* (or *communicate*) need not be actualised, but only occur in typical worlds

(31) $\llbracket\text{letter}\rrbracket = \begin{bmatrix} \text{ARG_STR} & = \lambda x.letter(x) \\ \text{TELIC} & = \lambda x.\lambda e.\forall w'\in R_{typ}.convey_{w'}(e)\wedge letter(x)\wedge CAUSE(e,x) \\ \text{AGENTIVE} & = \lambda x.\lambda y.\lambda e'.write(e',x,y)\wedge letter(x)\end{bmatrix}$

- Target: ⟦letter that p⟧ via coercion is interpreted as 'the set of letters that would in typical worlds convey that p':

(32) $\lambda x.\forall w'\in R_{typ}.\exists e.[letter(x)\wedge convey_{w'}(e)\wedge(CONT(e)=p)\wedge CAUSE(e,x)]$

- This will require the following covert operations:
  - RT to combine (TELIC(⟦letter⟧) with the ⟦that p⟧
  - Existential closure of the event argument to yield a set of letters

(33) $EC_{\text{int}} := \lambda R_{\langle\sigma,\langle\tau,t\rangle\rangle}.\lambda x_{\sigma}.\exists y_{\tau}.R(x)(y)$

Derivation: _letter that p_ via routinised coercion

(34) ⟦letter that p⟧ $\mapsto_{\text{coercion}} EC_{\text{int}}(RT(\text{TELIC}(\llbracket\text{letter}\rrbracket))(\llbracket\text{that p}\rrbracket)))$

Step-by-step:

(35) TELIC(⟦letter⟧) $= \lambda x.\lambda e.\forall w'_{\in R_{typ}}.convey_{w'}(e)\wedge letter(x)\wedge CAUSE(e,x)$

(36) ⟦that p⟧ $= \lambda e.[CONT(e)=p]$

(37) RT(35)(36) $= \lambda x.\lambda e.\forall w'_{\in R_{typ}}.convey_{w'}(e,y)\wedge letter(x)\wedge CAUSE(e,x)\wedge(CONT(e)=p)$

(38) EC_{int}(37) $= \lambda x.\exists e.\forall w'_{\in R_{typ}}.convey_{w'}(e,y)\wedge letter(x)\wedge CAUSE(e,x)\wedge(CONT(e)=p)$

8.3 Discussion

Problems:

1. Over generation: *start the letter* does not have a reading 'start conveying (with the letter)' interpretation
    - Possible fix: *start* is incompatibility with a modalised argument
      ?Alex started to possibly read.
2. The functionality of *Cont*
    - for *letter/email that p* the problem was that letters do not have unique propositional contents
    - Now we have: 'a letter conveying that p' – I.e., a *convey* eventuality with a unique contents p

Previous example:

> *Dear sir/madam,*
>
> *Your library book is one week overdue. Overdue library loans incur fines and so you need to pay a fine of €2.*

All of the following seem to be felicitous reports:

(39) a. I got an email that I need to pay a fine (of €2).
b. I got an email that my library book is overdue.
c. I got an email that overdue library loans incur fines.

- In order to uphold the functionality of *Cont*, we need to assume a multiplicity of eventualities associated with every letter, email etc.

$$\begin{aligned} e : \mathit{Cont}(e) &= [\![\text{Peter needs to pay a fine}]\!] \\ e' : \mathit{Cont}(e') &= [\![\text{Peter's library book is overdue}]\!] \\ e'' : \mathit{Cont}(e'') &= [\![\text{Overdue library loans incur fines}]\!] \\ e''' : \mathit{Cont}(e''') &= [\![\text{Peter needs to pay a fine}]\!] \wedge [\![\text{Peter's library book is overdue}]\!] \\ \ldots &= \ldots \end{aligned}$$

Tentative alternatives

- Weaken *Cont* to a relation (the same entity can have more than one proposition as contents)
- Let the relation be specified by the lexical entry of the noun (functional for nouns such as *belief*, relational for other nouns)

9 Conclusions and open questions

9.1 Take-home message

Seemingly, a routinised coercion-based account best accounts for the missive noun-NCC data

- But a number of issues need to be worked out

9.2 The cognitive cost of routinised coercion

There are alternative non-coerced constructions for many of the sentences in the dataset:

Verbal complementiser constructions with a PP:

(40) a. He wrote an email/a letter that he would appeal.
b. He wrote in an email/a letter that he would appeal.
c. He wrote that he would appeal in an email/a letter.

Adding a gerund:

(41) a. there had already been articles that she had a "cult"
b. there had already been articles saying that she had a "cult"

(42) a. You have to bring documents that I am Uighur.
b. You have to bring documents showing that I am Uighur.

- Why should speakers opt for the coerced constructions if they induce higher cognitive loads on hearers?

Further (indirect) evidence that routinised coercion does not carry a high cost

- Bott (2017): contextual information can be integrated during incremental sentence processing
- Further testing needed to see whether there are e.g., extra processing costs for *book that p* vs. *letter that p* depending on the supporting context
  - I.e., is lexically encoded 'Qualia' field information less costly to access and integrate than that provided by previous context

9.3 Open question: Alternatives to a type-driven account

- At least on the face of it, the use of NCCs with *article* and *document* are less frequent than with *email* and *letter*
- If this is confirmed, then an information theoretic approach to coercion (Delogu et al., 2017) may make different predictions from a type-driven coercion-based analysis.
  - For a type-driven approach, not obvious that *document* SHOWING *that p* should be harder to derive than *letter* CONVEYING *that p*
  - For an information theoretic approach, *document that p* would be predicted to generate more surprisal.

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A Data collection and cleaning

A.1 Initial gathering of *N that* sentences

For *N* in:

(43) article, book, document, email, letter, newspaper, website

a 'noisy' CQL search to pull sentences with *N that* constructions where *that* is not a subject relative pronoun (object relative pronouns not excluded at this stage):

(44) `<s/> containing [tag="NN|NNS"&lemma="N"] [word="that"] [tag="PP.*|NP.*|DT"]`[3]
Any sentence containing *N that DP*.

A random sample of 10k sentences per noun.

A.2 Distinguishing nominal from verbal complementisers

In a lot of the extracted NCC constructions, the N was the object of a *that*-complement taking verb

- *sign a document that p*, *read an article that p* etc.

However:

- Not all examples have a verb that takes a complementiser e.g. *there had already been *(articles) that she had a "cult,"* in (8a)
- Even when there is a complementiser-taking verb, unlike PP constructions, the complementiser is nominal here, not verbal. Evidence: if the noun was not critical in (45a), we would not expect (45b) to be infelicitous:

  (45) Constructed examples based on (8b):
  a. I read in an article that the rivers collected 3.8 billion gallons of water.
  b. ?I read a book that the rivers collected 3.8 billion gallons of water.

- There is no covert preposition insertion that makes (45a) felicitous, since (46b) is felicitous

  (46) a. I read in an article that the rivers collected 3.8 billion gallons of water.
  b. I read in a book that the rivers collected 3.8 billion gallons of water.

A.3 Examples of manually excluded constructions

Unclear/ungrammatical (here suspect a missing *know*)

(47) I'm new! but I wanted to let the one managing Marcelo's article that he has been transferred to French Ligue 1 "Le Mans Union Club 72".

Result constructions not detected in automatic filtering:

(48) I came across your site when I was looking for solutions about mouldy washing machine rubber, and, liked so much the articles that I'm still around reading them!

Relative pronouns not detected in automatic filtering (probable dependency parsing error):

[3] CQL tags and symbols. `[...]` is an expression in a string; `[...]*` is an optional expression in a string; `&` is conjunction; `|` is disjunction; `NN|NNS`: singular or plural common noun; `PP.*`: personal or genitive marked pronoun; `NP.*`: proper name or genitive marked proper name; `DT`: determiner; `IN`: preposition

(49) Las Vegas Sands insisted it needed approval from Macau authorities in order to give federal investigators documents that it turns out were in the U.S. all along.

Falsely tagged data (especially for *document*)

(50) It also removes some remnant control information about net.* and documents that the hierarchy is abandoned.

Verb-complement clause constructions (typically where N is in a PP):

(51) a. I have no idea who keeps putting on this article that Ivan IV was murdered, but the sources they use either are about something else entirely or state that the idea is very unlikely.
b. (Note the mention post article that Steve is former GCB writer, now PR privateer)

B Covert *say* insertion

say complementation (Grimshaw, 2015; Major, 2021)

- in English, such as *book* cannot be used to form subjects for progressive *say* constructions 52a or be used with goal arguments (see also discussion in Major, 2021, p. 27 and ch 2):

(52) Grimshaw 2015, p.87
a. ??The sign/poster/book/article was saying that the park was closed.
b. ??The sign/poster/book/article said to the tourists that the park was closed.

Perhaps covert *say* complementation underpins the use of NCC constructions with nouns such as *letter*?

- a covert reduced relative clause with *saying* or a covert full relative clause with *that says* is present as in (53a) and (53b).

(53) a. The letter (saying) that the park is closed is on the sideboard.
b. The letter (that says) that the park is closed is on the sideboard.

- Prediction: One of the following must hold to explain the contrast between *letter* and *book*:
  (A) Either *letter* is less restricted with *say* complementation than book
  (B) Or *book* is more restricted than *letter* in reduced relative or full relative clause constructions with *say*
- Prediction (A) is not borne out: compare (52a) and (52b) with (54a) and (54b)

(54) a. ??The letter was saying that the park is closed.
b. ??The letter said to the recipient that the park is closed.

- Prediction (B) is not borne out: compare (53a) and (53a) with (55a) and (55b)

(55) a. I remember the self-help book saying that often, other people don't even notice when anxious people are shaking. [ukWaC, 1768942]
b. Charles Darwin wrote a book that said that man was descended from apes [ukWaC, 1390374]

Conclusion:

- Covert *say* insertion is not the way to go